1 $T(n) = aT(n/b) + f(n)$

(a) $T(n) = 5T(n/4) + \log n$

$\rightarrow a=5,\ b=4,\ f(n) = \log n$

$n^{\log_4 5} > \log n$

$\rightarrow T(n) = O(n^{\log_4 5})$

(b) $T(n) = 16T(n/2) + n^4$

$\rightarrow a=16,\ b=2,\ f(n) = n^4$

$n^{\log_2 16} = n^4$

$\rightarrow T(n) = O(n^4 \log n)$

(c) $T(n) = 9T(n/3) + n^2 \log n$

$a=9,\ b=3,\ f(n) = n^2 \log n$

$n^2 \log n > n^{\log_3 9}$

$T(n) = O(n^2 (\log n)^2)$

2)

a) $T(n) = 3T(n/3) + O(1)$ ← Constant

$T(n) = \begin{cases} 3T(n/3) + c & n \geq 2 \\ c & n \leq 2 \end{cases}$

b) $T(n) = 3T(n/3) + c$

$T(n) = 3[T(n/3^2) + c] + c$

$T(n) = 3^3 T(n/3^3) + 3 \cdot c$

$T(n) = 3^k T\left(\frac{n}{3^k}\right) + c \rightarrow \frac{n}{3^k} = 1 \rightarrow 3^k = n$

$\Rightarrow k = \log_3 n$

$T(n) = n^{T(1)} + \log_3 n \, c \rightarrow O(n)$

(time complexity)

c) $m = bn/uc$

$T(n) = T\left(\frac{bn}{uc}\right) + T\left(\frac{bn}{uc}\right) + T\left(\frac{bn}{uc}\right) + c$

$\Rightarrow 3T\left(\frac{bn}{uc}\right) + c$

$T(n) = \begin{cases} 3T(bn/uc) + c & n \geq 2 \\ c & n \leq 2 \end{cases}$

d) $T(n) = 3T\left(\frac{bn}{uc}\right) + c$

$T(n) = aT(n/b) + \theta(n^k \log n) \rightarrow a = 3 \quad k = 0$

$b = uc/b \quad p = 0$

$\Rightarrow b^k = \left(\frac{4c}{b}\right)^0 = 1$

$T(n) = O(n^{\log_b a}) \Rightarrow T(n) = \Theta\left(n^{\log_{\frac{4c}{b}} 3}\right)$

3 method (Index, array, k)

make a loop function Index ~ k → i.

when array[i-1] > array[i]

Swap array[i] to array[i-1]

when array[k-i] < array[size-1-i]

Swap array[size-i] to array[s-1-i]

make a loop 1 to k

when k-i > i-1

↳ cycle with Index, array and k.

else → break

int count = 0;

make while → size > count

when array[i] > array[j]

array[count] = array[j]

j++

else

↳ array[count] = array[i]

i++

2)

a) T(n) =

b) T(n) =

T(n)

T(n) =

T(n) =

T(n) =

c)

m = b

T(n) =

=

d) T(n) =

T(n) =

=> b^k

T(n)

LLS, LRL,
S, LSS }

LR }

R, LLS,
RS, SLS,

RRL, RRS }

4

a) $\frac{n(n-1)}{2} \rightarrow \frac{n^2-n}{2} \rightarrow$ Big O $\Rightarrow O(n^2)$

b) Let $A = \{(1,2), (3,8), (6,11), (7,8)$

$A_1 \rightarrow \{(1,2), (3,8)\}$

$A_2 \rightarrow \{(6,11), (7,8)\}$

Array 1 $\rightarrow$ overlap : 0 , Array 2 $11-7+1 = 5$

max $(0,5) = 5$

$8-6+1 = 3 \rightarrow$ Should be answer

can't get correct answer enough

c) I don't know how to solve this questions

d) I don't know how to solve this questions

e) I don't know how to solve this questions.